

DeepAware: Using Experiential Deepfake Simulations to Enhance Cybersecurity Awareness in Older Adults

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Abstract

Deepfake scams, which use AI-generated audio or video to impersonate individuals, pose an increasing cybersecurity threat to older adults. Existing educational approaches present threats through generic examples, leaving learners to perceive scams as something that happens to others rather than to themselves. To address this gap, we conducted a formative study with five digital educators to identify design requirements, then developed DeepAware, a self-referential simulation platform that embeds participants' own faces and voices into deepfake scam scenarios. By making learners the target of simulated threats rather than passive observers, DeepAware aims to collapse the psychological distance between abstract warnings and personal vulnerability. A mixed-methods evaluation with 21 older adults found improvements in deepfake knowledge, threat perception, and coping confidence, though responses varied by prior familiarity. This work demonstrates the potential of self-referential simulation for cybersecurity education and offers design implications for future cybersecurity interventions.

CCS Concepts

• **Human-centered computing** → *Empirical studies in HCI*; • **Security and privacy** → *Social aspects of security and privacy*.

Keywords

Deepfake, Older Adults, Experiential Learning, Personalized Simulation, Cybersecurity Education

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1 Introduction

Deepfakes are AI-generated synthetic media that produce hyper-realistic impersonations of a person's face and voice [70, 103]. Leveraging deep learning models trained on facial imagery and voice recordings, these techniques can generate manipulated audiovisual content that convincingly mimics real individuals. As such tools become increasingly accessible and adaptable to everyday communication channels, with numerous open-source model variants readily available [44], deepfakes have emerged as a prominent tool for social engineering scams [43]. These scams often involve fraudsters impersonating family members or authority figures in distress to coerce victims into urgent financial transfers [30, 77]. Older adults, though more digitally active since the pandemic, remain particularly vulnerable due to limited awareness of emerging AI threats and lower confidence in digital security [9, 60]. Crucially, many older adults have never directly heard or encountered deepfake manipulation, making the threat feel abstract until real harm occurs.

Research on deepfakes has primarily focused on technical detection methods [38, 57, 65, 76, 101] and public-facing digital literacy initiatives [1, 31, 51, 111]. While these approaches are valuable, increasing knowledge alone does not always lead to accurate or timely judgment when scams unfold, as they often exploit emotional urgency and social trust. Technical tools are often embedded at the platform level and may be inaccessible in everyday decision-making, while general literacy campaigns tend to emphasize awareness rather than practice. Moreover, these efforts rarely consider the specific contexts and learning needs of older adults, who may face age-related challenges in memory, processing speed, and decision-making that affect their ability to evaluate and act on deceptive content [27, 41]. These challenges highlight the need for educational strategies that help older adults not only understand deepfake risks

conceptually but also experience how such fraud may personally target them.

Personalized simulation may offer a way to bridge this gap. By embedding users' own faces and voices into simulated scam scenarios, such interventions can collapse the psychological distance between "a scam that happens to others" and "a scam that could happen to me." Research on self-referential processing shows that seeing oneself in a risk context enhances attention, memory retention, and perceived threat [39]. When individuals experience a threat as personally relevant, they are more likely to retain information and sustain vigilance [40, 55]. Building on this theoretical perspective, we examine how personalized exposure may heighten risk perception in the context of deepfake scams, while also exploring whether it supports confidence in responding to such threats.

Guided by this perspective, we investigate the potential of personalized deepfake simulation for scam education among older adults:

- **RQ1:** What design considerations support meaningful and personally relevant deepfake scam education for older adults?
- **RQ2:** How do older adults perceive and evaluate a personalized simulation-based intervention that uses their own faces and voices?

To ground the intervention in practical teaching needs, we first conducted formative interviews with five digital educators who specialize in teaching older adults. They reported that deepfake instruction in current curricula is fragmented and heavily rule-oriented, offering limited emotional resonance and practical understanding. These insights informed our focus on creating experiential exposure to deepfake scenarios that would make threats feel personally relevant and emotionally salient as a central design requirement for scam education.

Building on these formative insights and prior research on experiential cybersecurity education [108, 109] and self-referential learning [39, 55], we designed and developed *DeepAware*, a web-based intervention that exposes learners to deepfake scam scenarios using their own face and voice. The system focuses on two deception strategies—identity theft and fake news—and presents simulations in which an attacker misuses the learner's identity to create emotional pressure. The intervention aims to heighten learners' awareness of their personal vulnerability to deepfake scams while equipping them with actionable strategies for protection.

To evaluate this design approach, we conducted a user study with 21 older adults in South Korea. Results showed improvements in deepfake knowledge, perceived threat, and coping efficacy. Participants emphasized that self-relevant simulations made scams feel immediate, while actionable coping strategies and review quizzes strengthened their confidence to respond. Overall evaluations were positive, though participants with prior deepfake familiarity found the content less novel, while those new to deepfakes cautioned that highly realistic scenarios might overwhelm some learners. These findings suggest that while personalized simulations can effectively enhance preparedness, their impact depends on calibrating scenario intensity to learners' baseline familiarity. On this basis, we derive design implications for cybersecurity education that emphasize personalization, adaptive scaffolding, and flexible support for older adults' diverse needs.

Together, our study demonstrates the potential of self-referential deepfake simulation to address the gaps in older adults' education on deepfake scams. Building on this foundation, our work makes the following contributions:

- (1) Identifying design requirements for deepfake scam education tailored to older adults through a formative study with five digital educators.
- (2) Designing and implementing *DeepAware*, a personalized deepfake simulation system to investigate how embedding users' own faces and voices into simulated scenarios can support deepfake scam education for older adults.
- (3) Demonstrating through evaluating with 21 older adults that personalized simulations strengthen perceived threat, coping efficacy, and engagement, with design implications for experiential deepfake education.

2 Related Work

2.1 Older Adults' Vulnerabilities to Digital Fraud and Deepfakes

Deepfakes—hyper-realistic synthetic audio, video, or images generated with deep learning [70, 103]—are increasingly leveraged in digital fraud schemes targeting older adults. When combined with social engineering tactics [43], these technologies enable convincing impersonations in schemes such as grandparent scams, romance scams, investment scams, tech support fraud, and government impersonation scams [30, 77]. The "grandparent scam," for instance, now uses synthetic voices and video impersonations to mimic a family member in distress, dramatically heightening credibility and making fraudulent solicitations appear authentic [30]. As these manipulative techniques permeate everyday communication, they challenge the relational familiarity that older adults often rely on for authenticity judgments [84].

Empirical evidence indicates that older adults face particular challenges in detecting deepfake deception. Studies consistently show that people struggle to distinguish AI-generated media from authentic content, performing only marginally better than guessing [23, 26], and that older adults perform consistently worse than younger adults [47, 78, 105]. This detection difficulty arises not only from a lack of awareness but also from deeper vulnerabilities rooted in cognitive, emotional, and social factors. Age-related changes in processing speed and working memory can impair rapid evaluation of suspicious requests [41], while perceptual and cognitive declines hinder the application of detection heuristics under stress [78]. Beyond cognition, stress, uncertainty, and fear often drive avoidance rather than protective action [73]. Additionally, low self-efficacy could further reduce the willingness to question suspicious requests or attempt verification [74]. Reliance on relational trust makes family impersonation scams particularly effective against older adults [91, 109], while economic pressures and perceived responsibility to protect family members can override rational risk assessment [50].

Existing responses to deepfake threats, while valuable, only partially address the challenges older adults face in practice. Technical detectors, despite rapid advances, still struggle to generalize, as models trained on academic datasets exhibit substantial performance drops when evaluated on in-the-wild or user-generated content [3, 38, 82]. Model-driven techniques rely on complex pipelines

such as eye-blink detection [63], head-pose analysis [67, 106], and audio–visual consistency checks [65, 110], all of which require expertise or interpretation skills that most older adults could not practically employ. Training users to detect such artifacts also shows limited real-world utility given the perceptual and cognitive challenges noted above. Institutional education focuses on conceptual awareness (e.g., fraud typologies and warning signs) rather than preparing individuals to respond during emotionally heightened scam situations [1]. Consequently, many older adults lack practiced strategies for recognizing or resisting deepfake-based deception in real-time encounters [33, 36].

2.2 Experiential Learning: From Knowledge to Protective Action

Most cybersecurity education for older adults emphasizes conceptual understanding through videos, chatbots, and games [18, 68, 97], yet research suggests that knowledge alone does not reliably produce protective behavior. Protection Motivation Theory (PMT) provides a framework for understanding this gap [69, 86]. PMT posits that protective behavior arises from two cognitive appraisals: threat appraisal—perceiving the threat as personally relevant (vulnerability and severity)—and coping appraisal—believing one can effectively respond (self-efficacy and response efficacy). Knowledge-based instruction may raise awareness of deepfakes (contributing to perceived severity), but often fails to establish personal vulnerability—the sense that “this could happen to me”—or build confidence in one’s ability to respond effectively [23, 26]. When individuals perceive a threat as serious but lack confidence in their ability to act, they often resort to avoidance or disengagement rather than protective behavior [6, 46].

Experiential learning offers a pathway to strengthen both threat and coping appraisals by enabling learners to encounter threats and practice responses in controlled environments. Rather than presenting information passively, experiential approaches simulate realistic scenarios that make abstract threats feel immediate (strengthening threat appraisal) while providing practice opportunities that build confidence in protective actions (strengthening coping appraisal) [10, 56]. Research across domains demonstrates this effectiveness: phishing role-plays improve detection and reporting [89]; cyber-range simulations support behavioral transfer [59]; experiential dark-pattern training heightens threat salience [108]. These studies show that experiential learning can transform awareness into actionable behavior by addressing both cognitive pathways identified by PMT.

However, existing experiential approaches could have limitations when applied to older adults and deepfake scams. Most implementations target younger populations and rarely accommodate older adults’ diverse learning needs [59, 92]. More critically, current approaches have not adequately addressed the challenge of perceived vulnerability. While experiential simulations effectively build coping skills through practice, they typically use generic scenarios (e.g., “imagine you receive a suspicious email”) that maintain psychological distance between the learner and the threat.

Prior work on tailored security messaging shows that personalization can strengthen perceived relevance [34, 53, 74], but these approaches often focus on adapting message content or delivery

rather than situating learners as direct targets within threat scenarios. For deepfake scams—where emotional manipulation combines with technical sophistication—this limitation is particularly consequential. Learners may understand that deepfakes exist, practice detection strategies, and even believe scams are serious, yet still fail to internalize that they themselves could be targeted until experiencing an actual attack.

2.3 Self-Referential Personalization: Making Threats Personally Tangible

Reducing the psychological distance between learners and threats requires approaches that make risks feel personally relevant rather than abstract. The self-reference effect—a well-established phenomenon in cognitive psychology—demonstrates that information tied to one’s own identity undergoes deeper processing and produces stronger recall than information about others [39, 87]. When people encounter self-relevant information, they engage more elaborative encoding, connecting new content to existing self-knowledge and personal experiences [40]. This effect is particularly pronounced for older adults, who show enhanced memory performance for self-relevant compared to other-relevant material [39, 55].

Recent work demonstrates that self-resemblance enhances engagement and learning across domains. Incorporating learners’ own faces or voices into educational materials has been shown to increase attention, reduce anxiety with new technologies, and strengthen emotional connection [29, 37, 54]. In health interventions, increasing self-identification with a virtual scenario has been shown to elevate self-efficacy and intensify the experiential impact of the simulation [49]. In virtual training, self-resembling avatars have been shown to deepen engagement and social presence, enhancing users’ sense of co-presence and believability compared to generic representations [37]. These findings suggest that self-referential content can bridge the psychological distance between abstract information and personal relevance, strengthening both the perceived immediacy of threats and learners’ confidence in responding to them.

In the context of deepfake scams, self-referential simulation presents a distinctive opportunity. Rather than describing how deepfakes might manipulate “someone like you,” interventions could show older adults their own faces and voices embedded in realistic scam scenarios. This direct exposure may strengthen both threat recognition and coping responses in ways that generic examples cannot. We developed DeepAware to explore this approach—a platform that generates personalized deepfake simulations using participants’ own photos and voice recordings. Through interviews with digital literacy educators and evaluation sessions with older adults, we examine how to design such interventions effectively and how learners experience this form of self-referential education.

3 Formative Study

To explore how deepfake-related risks can be effectively taught to older adults, we first sought formative insights from educators who specialize in this population. These digital educators are not general instructors but practitioners with extensive expertise in tailoring instruction to the needs, challenges, and learning styles of older adults. They are recognized as central actors in digital inclusion,

Table 1: Demographic and professional background of digital educators interviewed in the formative study, including their teaching experience and major course areas with older adults.

ID	Age	Years of Experience	Class size (avg.)	Major Teaching Courses
E1	55	7	20–40	Smartphone utilization, smartphone video editing
E2	57	10	10–15	Smartphone utilization, YouTube content creation
E3	54	15	20–40	Smartphone utilization, AI, GPT application
E4	54	11	20–30	Smartphone utilization, SNS marketing
E5	56	9	20–40	Computer skills, Digital literacy, Online safety

trained to address the learning needs of older adults and frequently serve as intermediaries between technology and those who struggle to access it [33, 36]. Because they work with diverse learners over extended periods, educators can identify recurring patterns and challenges that individual older adults may not articulate in one-time interviews. Prior work has similarly engaged digital educators to understand how older adults learn general digital skills [94]. Building on this perspective, we recruited educators as key informants to better understand current teaching practices, identify unmet needs, and inform the design of a personalized deepfake education system that we subsequently evaluated with older adult learners.

3.1 Method

We interviewed five digital educators, all highly experienced practitioners with more than five years of teaching older adults in community education centers across South Korea. Participants were recruited through purposive sampling via direct outreach, with four from community centers in Seoul and one from a rural area. Table 1 summarizes participant backgrounds.

Each interview followed a semi-structured interview protocol and was conducted in person or via video conferencing, lasting 60–90 minutes. Before the session, participants received an explanation of the study’s purpose and procedures and provided informed consent for participation and audio recording. The protocol explored three main areas: (1) current practices and challenges in teaching older adults, (2) approaches to explaining digital safety and emerging risks such as deepfakes, and (3) educators’ observations of how older adults respond to different forms of instruction and warnings. The complete interview questions are included in the Appendix Appendix A.

All interviews were transcribed and analyzed using thematic analysis[15]. Following an inductive approach, three researchers independently conducted open coding to identify initial codes. Codes were iteratively compared, merged, and refined through discussion. Emerging patterns were organized into candidate themes, reviewed against the data for coherence and distinctiveness, and finalized through consensus. This process yielded four central themes that capture both key challenges and effective practices for teaching older adults about digital risks.

3.2 Findings

3.2.1 Finding #1: Deepfake education remains superficial due to structural constraints. Educators frequently described older adults’ strong motivation to engage with digital learning,

often linked to family connections and a desire for independence. E3 explained, “*After COVID-19, there were so many things you had to do with your phone. Many learners said their children encouraged them to join the class instead of just helping them at home.*” Similarly, E4 observed, “*Some participants say they feel bad constantly asking their children for help, so they want to learn to do things themselves.*”

At the same time, educators reported that many older adults remained anxious about basic digital operations, particularly tasks that required logins or account management. E1 reflected, “*They’re not as afraid of smartphones as before (before COVID-19), but some still worry about pressing the wrong button and causing damage.*” To address this concern, classes often centered on repetitive practice with everyday functions such as texting or creating group chat-rooms. Educators noted that this emphasis helped build confidence but left limited room for more advanced or safety-related topics.

Several also pointed to constraints in their own expertise and in institutional structures. Some admitted they were not confident in teaching how to detect or respond to deepfakes. As E1 said, “*Even educators do not fully understand deepfake detection methods, so we cannot adequately cover them in class.*” Program structures reinforced these limits, since safety topics were often confined to occasional special lectures rather than continuous instruction. E1 explained, “*Digital crime prevention is important, but we usually cover it only once a quarter as a supplement rather than a regular course.*” When digital safety was addressed, educators said the content often focused on simple heuristics—avoiding unknown calls or confirming requests with family members. Yet they also observed that these approaches had limitations. As E2 noted, “*In real emergency situations, older adults may panic and fail to perform the verification steps effectively.*”

3.2.2 Finding #2: Self-relevant contexts elicit stronger engagement. Given that current approaches remain abstract and lack personal salience, educators emphasized the need for content that makes threats feel personally urgent. They noted that identity theft and fake news drew particularly strong reactions when learners had witnessed similar incidents in their communities or heard about them from acquaintances. E3 remarked, “*When they hear someone they know got scammed, that’s when it really hits them. They would then start asking if this could happen to me as well.*”

Building on this sensitivity, educators stressed that educational materials must tap into contexts that feel directly connected to learners’ lives. They suggested that stories involving family or close relationships, in particular, would create a sense of immediacy that general explanations could not. E2 remarked, “*When the stories are about family or friends, they really pay attention.*”

However, educators argued that family-centered narratives alone might not be sufficient to translate concern into protective action. Several anticipated that directly presenting learners' own faces or voices would make the threat more visceral. E5 explained, "*On site, you need a shock effect. When they see their own photo synthesized, they understand naturally.*" Similarly, E1 noted, "*When they see crime-related videos, they think it's interesting but when they see their own face in it, the reaction will be completely different.*" Based on these observations, educators viewed self-referential content—embedding learners' own identities into realistic scenarios—as potentially essential for helping older adults internalize deepfake threats as both believable and personally consequential.

3.2.3 Finding #3: Progressive scaffolding builds confidence while reducing anxiety. At the same time, educators cautioned that self-referential content could overwhelm learners if introduced too abruptly. They emphasized the importance of progressive exposure, noting that many older adults are often unfamiliar with the term "deepfake" itself. E3 explained that guided rehearsal in low-pressure settings would allow learners to realize scams "*could happen to them*" while maintaining a sense of control, whereas abrupt exposure to high-stakes scenarios can trigger avoidance. Several educators anticipated that presenting personalized deepfake scenarios without preparation might create anxiety rather than facilitate learning.

Drawing these observations together, educators suggested that effective deepfake education may benefit from scaffolding learning progressively—beginning with conceptual grounding about what deepfakes are and how they work, then introducing example cases, and only later presenting personalized simulations that embed learners' own identities.

3.2.4 Finding #4: Concrete coping strategies are more actionable than abstract warnings. Beyond structuring exposure, educators stressed the importance of providing learners with actionable tools. They noted that coping strategies felt most useful to older adults when they were simple, concrete, and immediately applicable, particularly for deepfake education, where the underlying AI technology can feel abstract and intimidating. Educators contrasted different instructional approaches: telling older adults to "don't trust everything online" felt vague and paralyzing, while teaching a specific verification question (e.g., "ask something only the real person would know") felt actionable. They observed that learners in their classes often responded positively to concrete steps with phrases like "*I can do this right away*" (E4).

3.3 Design Requirements

The formative interviews reveal a central tension in deepfake education for older adults: learning experience must be personally relevant enough to convey urgency and relevance, yet carefully structured to avoid overwhelming learners or triggering disengagement. To address this tension, we established four design requirements that integrate insights from both the formative study and prior research:

DR #1: Situate the learner as a direct target of the simulated threat (Findings #1, #2). Both educators and prior research highlight the importance of self-referential content, educators anticipated

that seeing one's own face would create a "shock effect," while the self-reference effect demonstrates that self-relevant information undergoes deeper cognitive processing [39, 87]. We implemented this by having learners upload their own photos and record their voices during onboarding, which are then synthesized into realistic deepfake scam scenarios to strengthen perceived threat severity and personal vulnerability.

DR #2: Ground scenarios in high-salience threat types (Finding #2). Educators emphasized that identity theft and impersonation contexts resonated most strongly with older adults who had witnessed similar incidents. Building on this insight and recognizing that fake news represents one of the most prevalent forms of deepfake misuse [103], we designed two scenario types: identity theft (where the learner's face and voice are used to impersonate them to family members) and fake news (where the learner appears in fabricated news content). These contexts were intended to support threat appraisal by combining threats that are personally familiar with those widely recognized as common deepfake applications.

DR #3: Scaffold learning through progressive exposure (Finding #3). Educators cautioned against abrupt exposure to self-referential content, noting that it could trigger anxiety rather than learning—a concern supported by research on experiential learning, which shows that progressive scaffolding builds both comprehension and confidence-building [89, 108]. Therefore, we structured the learning experience in three sequential stages: first, conceptual education about deepfake technology; second, presentation of example deepfake cases; and third, immersive simulation using learners' own identities.

DR #4: Provide concrete, actionable coping strategies (Finding #4). Educators stressed that concrete, actionable strategies felt more useful than vague warnings, aligning with prior work on digital literacy for older adults [17]. To address this, we designed the system to teach specific verification techniques after the personalized simulation stage. These include asking knowledge-based questions or checking through alternative channels. This approach was designed to support the emphasis of Protection Motivation Theory on response efficacy and self-efficacy [69, 86], enabling learners to understand how they can effectively respond to deepfake threats.

4 DeepAware

Guided by the design requirements established in Section 3.3, we developed *DeepAware* to probe how older adults perceive and evaluate self-referential deepfake education. The platform embeds users' own photos and voices into simulations of identity theft and fake news, aiming to help older adults develop both threat awareness and coping confidence against deepfake scams.

4.1 System Component

DeepAware operationalizes these design requirements through three interconnected features: self-relevant simulations (**DR #1, 2**), scaffolded learning stages (**DR #3, 4**), and a self-resembling guide (**DR #1, 3**).

Self-Relevant Simulations. To heighten perceived threat through self-relevant content (**DR #1**) and ground learning in familiar contexts (**DR #2**), *DeepAware* provides two simulation modules—**Identity Theft** and **Fake News**—that embed learners' own photos and voices

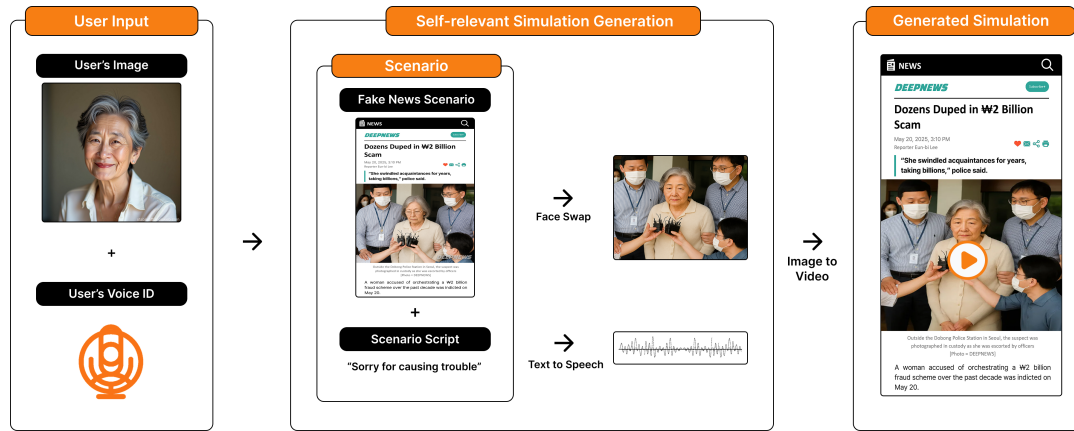


Figure 1: Process of the self-relevant simulation generation. This figure illustrates how a user's photo and custom voice ID are taken as input. The system applies face swap and text-to-speech techniques to embed the user's face and voice into a fake news scenario. The output is a video of the user's face embedded in fake news, delivering the scripted line. For demonstration purposes, the images shown in this paper use an AI-generated face; no real participant's likeness is depicted.

into generated media (see Fig. 1). These threat types were selected based on educators' observations that identity theft resonated strongly with older adults, and recognition that fake news represents one of the most prevalent forms of deepfake misuse. Scenario scripts were developed based on documented deepfake scam cases from law enforcement advisories [30, 79] and news reports [52, 62, 71], selecting scenarios that educators identified as plausible and emotionally resonant for Korean older adults. Each module introduces two sequential scenarios that follow a gradual arousal structure to help learners adapt to increasing emotional intensity.

- **Identity Theft Module:** This module introduces the misuse of the learner's cloned voice in communication contexts. The first scenario presents a situation where the learner's cloned voice encourages a friend to explore an investment opportunity, introducing voice-based impersonation without immediate time pressure. The second scenario heightens emotional intensity by using the learner's cloned voice to impersonate them in a video call with a family member, urging an immediate money transfer after a fabricated car accident.
- **Fake News Module:** In this module, learners' faces and voices appear in manipulated news content. The first scenario presents a lower-pressure positive situation where the learner appears as a lottery winner. The second scenario escalates to a higher-pressure situation that falsely reports the learner being accused of fraud in a fabricated news article.

Scaffolded Learning Stages. Each module follows a five-stage learning flow (Fig. 2): 1) concept introduction → 2) real-world examples → 3) self-relevant simulation → 4) coping strategies → 5) review quiz. This progression aims to support incremental comprehension (DR #3) by moving learners from foundational concepts to personalized experience, and to build coping efficacy (DR #4) by pairing threat exposure with actionable protective strategies.

- **1) Concept introduction:** The system explains the specific threat type, what identity theft or fake news deepfakes are

and how they work, providing a clear foundation that orients learners to the current module's focus.

- **2) Real-world examples:** Building on this foundation, learners view news clips of documented deepfake incidents to see how these threats have affected real victims. These curated media reports help learners connect abstract concepts to recognizable situations before encountering personalized simulations.
- **3) Self-relevant simulation:** Learners then experience the personalized deepfake scenarios described above, applying the threat patterns they observed in real-world cases to their own likeness. In the Identity Theft module, learners encounter their cloned voice being used to solicit an investment from a friend and to request emergency funds from a family member. In the Fake News module, learners see their face appearing in fabricated news reports as a lottery winner and later as someone accused of fraud.
- **4) Coping strategies:** Following each simulation, the system provides concrete, actionable guidance to channel heightened threat awareness into protective behavior. These strategies were synthesized from publicly available educational materials on deepfake detection and scam prevention [2, 35]. For the Identity Theft module, learners are taught practical verification methods: establishing family-specific code words to verify identity, being cautious about sharing photos online that could be used for impersonation, and saving contacts by name rather than relationship labels (e.g., "daughter" or "son") to reduce vulnerability to caller ID spoofing. For the Fake News module, the system guides learners through detection cues, such as examining facial inconsistencies, observing eye blinking patterns, checking lip-sync accuracy between mouth movements and audio, and noting unnatural skin texture.
- **5) Review quiz:** Each module concludes with a short quiz that prompts learners to recall key concepts and coping strategies. Prior research suggests that quiz-based retrieval

practice supports long-term retention, particularly for older adults when immediate feedback is provided [8, 85]. Our quizzes incorporate audio feedback to support self-paced consolidation before proceeding to the next module.

Self-Resembling Guide. To guide learners through this scaffolded progression, an animated self-resembling character generated from the learner’s photo and voice serves as a continuous companion. This design choice intends to extend self-referential processing beyond the threat simulations. Prior work suggests that self-resemblance can heighten engagement and reduce feelings of eeriness, helping older adults feel present in the training experience [29, 37, 54]. This character serves as a familiar and consistent guide, actively leading learners through each stage of the modules—introducing concepts, explaining the shift between module stages, presenting coping strategies, and offering feedback in the learner’s own voice. By maintaining a continuous presence, the character is intended to reduce anxiety and sustain engagement. The system also incorporates replayable audio guidance and simplified navigation, allowing learners to progress through the material at a comfortable pace without feeling rushed.

4.2 Implementation

DeepAware was developed as a web-based platform with a React frontend and FastAPI backend. The frontend manages the interface and module sequencing, while the backend coordinates AI-based content generation. To create immersive self-relevant scenarios, we evaluated several deepfake technologies and adopted the Akool API [5] because of its flexibility and reliability in generating video content tailored to our use cases.

Personalization was achieved by combining multiple AI services. Animated self-resembling characters were generated from user-uploaded photos using the OpenAI API [80], while natural speech was produced through ElevenLabs [28]. All media and user data were temporarily stored on AWS S3 [7] and Supabase [95] for processing and securely deleted after session completion. The final system was deployed on Vercel [99] to ensure stable access through the web.

The system was iteratively refined prior to formal evaluation. Four pilot sessions with older adult participants and six follow-up meetings with educators guided improvements in usability, content clarity, and generation performance. These sessions led to optimizations in UI flow, faster scenario generation, and more natural behavior of the animated self-resembling character, resulting in a smoother and more engaging learning experience.

4.3 User Scenario

This section illustrates how *DeepAware* is experienced from a learner’s perspective.

Before beginning, the user uploads a photo and records one minute of speech by answering daily questions. From this input, the system generates an animated self-resembling character that guides learners through each stage of the program, delivering instructions and feedback in their own voice.

In the *Preparation stage*, the learner watches a short video introducing deepfakes and then completes a brief quiz distinguishing real from AI-generated media. Immediate audio feedback clarifies

misconceptions, ensuring that learners start with a shared baseline of understanding and setting the stage for deeper engagement.

In the *Main learning stage*, the learner selects either the Fake News or Identity Theft module and proceeds through the scaffolded learning stages shown in Fig. 2. For example, in the Fake News module, the user sees a manipulated news clip falsely announcing them as a lottery winner, voiced by a cloned version of their own speech. After this experience, the system teaches key heuristics for deepfake detection and encourages fact-checking behaviors—such as verifying information through trusted sources or calling back using known numbers—before presenting short review questions with voice feedback. At the end of each module, the user may either finish the program or return to complete the other module, allowing learners to flexibly engage with both types of scams.

Finally, in the *Wrap-up stage*, the system displays a completion message and provides positive reinforcement to encourage the application of coping strategies in daily life.

5 User Evaluation

The goal of the user evaluation was to understand how older adults experience and evaluate self-referential deepfake education. Specifically, we examined their comprehension of deepfake threats, perceptions of threat appraisal, and coping efficacy when encountering simulations using their own faces and voices. Given the novelty of this approach for older adult populations, we adopted an exploratory single-group pre-post design to investigate whether personalized, experiential learning shows promise as an educational strategy.

5.1 Recruitment & Participants

We recruited 21 older adults (Mean age = 65.8, range = 61–80; 15 female, 6 male) through snowball sampling, drawing primarily on referrals from digital educators in the formative study and their local community networks. We adopted this recruitment strategy for two reasons. First, the study required participants to share personal photos and voice recordings for deepfake generation—a request that raised understandable privacy concerns when presented through impersonal channels such as online announcements. Establishing trust through known intermediaries proved essential for recruitment. Second, we targeted older adults who were already engaged with digital learning, as the web-based platform requires a baseline familiarity with smartphone interfaces and multimedia content, allowing us to assess the intervention’s potential without confounding learning outcomes with basic usability barriers. This recruitment approach resulted in a sample that skews toward digitally engaged older adults, predominantly female (15 of 21 participants), reflecting the composition of community-based digital education programs in South Korea [48]. All participants were native Korean speakers residing in urban areas of South Korea, comprising 12 retired individuals and 9 who were still engaged in part-time or full-time work. Detailed participant characteristics, including age, occupation, and device usage, are provided in Table 4 in the Appendix.

Participants were eligible if they actively used smartphones for daily communication, were familiar with messaging apps, and could

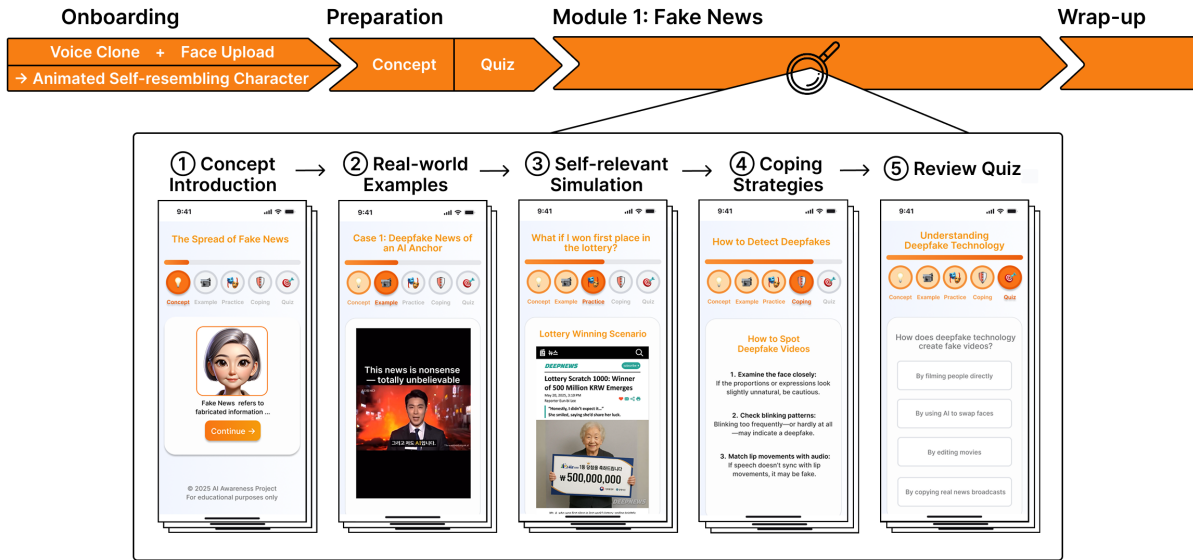


Figure 2: Scaffolded learning flow in *DeepAware*. The figure depicts the full instructional sequence, beginning with *Onboarding* (photo and voice upload to generate an animated self-resembling character) and *Preparation* (deepfake introduction and baseline quiz). The *Main Learning Stage* follows a five-step scaffolded flow: (1) *Concept Introduction*, where the character explains how deepfakes spread; (2) *Real-world Examples*, which show authentic cases of scams; (3) *Self-relevant Simulation*, where the learner's own photo and voice are embedded into a manipulated scenario; (4) *Coping Strategies*, which teach cues for detecting deepfakes and fact-checking behaviors; and (5) *Review Quiz*, which reinforces key concepts with immediate feedback. The sequence concludes with a *Wrap-up* stage that provides positive reinforcement and encourages transfer of strategies to daily life.

comfortably engage with multimedia content on screen. Before beginning the user study, the study's purpose and procedures were fully explained, and written informed consent was obtained. The study was approved by the Institutional Review Board (IRB) of the university where the study was hosted. Each study lasted approximately 90 minutes, and participants received 30,000 KRW (approximately 23 USD) for their participation.

5.2 Study Procedure

Once participants confirmed their availability and scheduled their study, we provided a brief overview of the study's purpose and instructed them to prepare a smartphone with internet access for the web-based learning experience. Studies were conducted either in person or remotely via Zoom, depending on participant preference. For the remote study, participants were guided to share their screens to ensure smooth navigation throughout the study.

At the start of each session, the researcher explained the study's objectives, obtained written consent for audio recording and data handling, and provided a short orientation on how to access and use *DeepAware*. Participants then completed a pre-survey, which collected demographic information and measured digital literacy [22, 100] and AI literacy [19, 66, 97]. Also, we measured participants' baseline deepfake knowledge, perceived threat (vulnerability and severity), and coping efficacy (self-efficacy and response efficacy). They then engaged in approximately 30 minutes of self-paced exploration of the *DeepAware* modules. During this time, they were

encouraged to verbalize their thoughts and reactions using a think-aloud protocol [13], while the researcher observed and took notes on both verbal and non-verbal responses.

After completing the modules, participants filled out a post-survey that re-assessed deepfake knowledge, perceived threat, and coping efficacy to capture changes, and additionally measured learning experience through usability, satisfaction, perceived usefulness, deepfake realism, and character immersion scales. Finally, each participant took part in a 20-minute semi-structured interview reflecting on their learning experience, perceptions of deepfake threats, and coping efficacy.

5.3 Measures

To examine whether *DeepAware* provides promising support to older adults' protective behavior, we assessed both *learning outcomes* and *learning experience*. Learning outcomes included *deepfake knowledge* and four Protection Motivation Theory constructs: perceived threat (*perceived vulnerability*, *perceived severity*), coping efficacy (*self-efficacy*, and *response efficacy*) [69, 86]. To understand how older adults engaged with the personalized, simulation-based learning, we also assessed learning experience, including *usability*, *satisfaction*, *perceived usefulness*, *deepfake realism*, and *character immersion*. All survey items were rated on a 5-point Likert scale (1 = strongly disagree, 5 = strongly agree). Subscale scores were computed by averaging the items. Unless otherwise noted, measures were collected both before and after the learning modules.

The complete list of survey items for PMT constructs and deepfake knowledge is provided in Appendix Appendix D, and learning experience items are provided in Appendix Appendix C.

5.3.1 Knowledge of Deepfake. Participants' conceptual understanding of deepfakes was measured with three items. Items assessed whether participants knew the concept and definition of deepfakes, recognized that the technology can manipulate faces and voices, and understood its increasing sophistication. These items were grounded in AI literacy frameworks that emphasize conceptual and functional understanding [19, 97]. The items formed a reliable scale (Cronbach's $\alpha = .89$).

5.3.2 Perceived Threat. Perceived threat was measured through vulnerability and severity, following Protection Motivation Theory [14, 46, 64]. **Perceived vulnerability** was assessed with three items such as "I am aware that someone could send a deepfake video to deceive me or my family." These items created a consistent scale (Cronbach's $\alpha = .74$). **Perceived severity** was assessed with two items including "I believe deepfakes can contribute to the spread of false information in society." The two items created an acceptable scale consistent (Cronbach's $\alpha = .62$).

5.3.3 Coping Efficacy. Coping efficacy was measured through self-efficacy and response efficacy. **Self-efficacy** was assessed with two items, adapted from Bandura's framework and AI literacy research [11, 18], asking participants to reflect on their confidence in recognizing manipulation, such as "I can tell if the news or videos I see online have been manipulated" and "I can tell whether the person I interact with online is human or AI." The two items were internally consistent (Cronbach's $\alpha = .83$). **Response efficacy** was assessed with three items, adapted from PMT-based behavioral research [21, 75], measuring beliefs about the effectiveness of protective strategies, including "When I see shocking or surprising videos, I will stop and think before believing them." These items formed a reliable scale (Cronbach's $\alpha = .76$).

5.3.4 Learning Experience. Beyond PMT-based learning outcomes, we assessed participants' subjective experience with DeepAware. First, we assessed the system's usability, satisfaction, and perceived usefulness, which are critical factors for the adoption of educational technologies by older adults [61, 90]. Additionally, we measured the realism of the deepfake simulations and participants' sense of immersion with the self-resembling character to understand how self-referential personalization shaped engagement [37, 54, 112].

Usability was measured using the ten items of the System Usability Scale (SUS) [16]. *Overall Satisfaction* was assessed with three items examining the coherence of the module flow and appropriateness of the activity design, which together formed a consistent scale (Cronbach's $\alpha = .77$). *Perceived usefulness* was assessed with two items capturing the practical value of DeepAware for recognizing risks such as identity theft (Cronbach's $\alpha = .86$) [24].

Deepfake Realism was assessed with a single item measuring how closely the deepfake simulations resembled the participant, using the scale from prior work on self-resemblance and credibility [37, 54]. Finally, *Character Immersion* was assessed with two items measuring the extent to which participants felt engaged with their animated self-resembling character (Cronbach's $\alpha = .91$) [112]. All

survey items were pilot tested with older adults to ensure clarity and appropriateness.

5.4 Semi-structured Interview

After the post-survey, participants took part in a 20-minute semi-structured interview to capture aspects of the learning experience. The interview explored four areas: emotional engagement, changes in perceived threat, applicability of coping strategies, and feedback on system design. The full interview questions is provided in Appendix E.

To probe emotional engagement, participants were asked how they felt when their own face and voice appeared in the simulation and how they experienced the animated self-resembling character. To examine shifts in perceived threat, we asked whether their views of deepfake scams had changed after using the system and which moments felt most realistic or threatening. Coping efficacy was explored by inviting participants to reflect on how their responses to suspicious content might differ after the program, or whether they had become newly aware of protective actions. Finally, participants reflected on system usability and instructional design by considering whether they would revisit the content, share it with family, or suggest improvements.

5.5 Analysis

We adopted a mixed-methods approach to analyze both quantitative and qualitative data, in order to capture not only measurable changes in knowledge and efficacy but also the lived reactions of participants.

5.5.1 Quantitative Analysis. Pre- and post-survey data were analyzed using the Wilcoxon Signed-Rank Test, a nonparametric method suitable for small samples and ordinal data [104]. Effect sizes were computed as $r = Z/\sqrt{N}$, with values of 0.1, 0.3, and 0.5 interpreted as small, medium, and large effects, respectively. For post-survey measures including *Usability*, *Satisfaction*, *Perceived Usefulness*, *Deepfake Realism*, and *Character Immersion*, we reported descriptive statistics to summarize participant responses.

5.5.2 Qualitative Analysis. Qualitative data were drawn from transcribed interviews and think-aloud verbalizations during the session. We conducted an open coding process in which researchers segmented transcripts into meaningful units and identified recurring patterns. Through iterative discussions, the team refined a shared codebook and organized findings into four dimensions corresponding to our survey constructs: *Shifts in Knowledge of Deepfake*, *Shifts in Perceived Threat*, *Shifts in Coping Efficacy*, and participants' *Learning Experience* (covering usability, perceived usefulness, deepfake realism, and character immersion).

5.6 Ethical Considerations

To protect participant privacy in all published materials, the figures in this paper utilize AI-generated images instead of actual participant images. While informed consent and IRB approval were obtained prior to the study, we took further steps to protect participant privacy and emotional well-being. All uploaded images and voice recordings used to generate personalized simulations were deleted immediately after the session. Media processed by

Table 2: Descriptive statistics for post-program learning experience measures ($N = 21$). All measures used 5-point Likert scales.

Measure	Items	α	M	SD	Median	IQR
Satisfaction	3	.77	4.55	0.61	5.00	0.70
Perceived Usefulness	2	.86	4.48	0.49	4.50	1.00
Deepfake Realism	1	–	3.21	1.02	3.00	1.00
Character Immersion	2	.91	4.26	1.03	4.50	1.00

– indicates single-item measure.

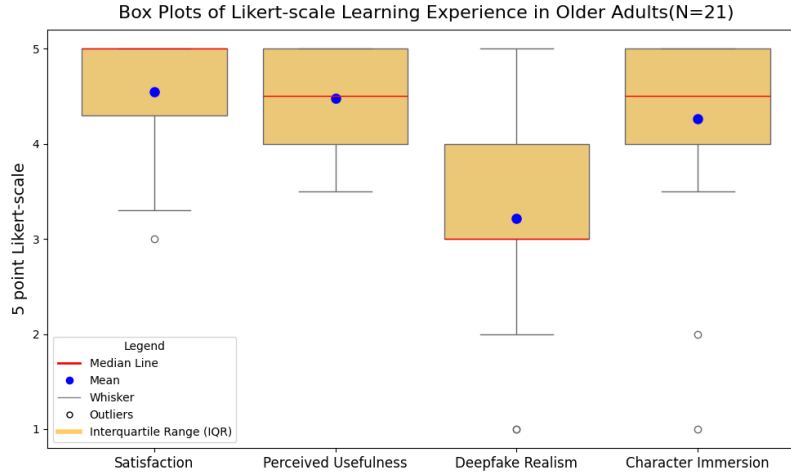


Figure 3: Boxplots of post-survey ratings on four dimensions of the learning experience: Satisfaction, Perceived Usefulness, Deepfake Realism, and Character Immersion. Ratings were given on a 5-point Likert scale (1 = Strongly Disagree, 5 = Strongly Agree). Overall, satisfaction and usefulness received the highest ratings, realism was evaluated more moderately, and immersion was rated positively with some variability.

third-party services (e.g., Akool and ElevenLabs) was used solely for real-time generation and was permanently deleted from those services upon completion of the session. Furthermore, because the deepfake threat scenarios could evoke stress or discomfort, we implemented multiple safeguards. Participants were given a preview of the scenario content and informed that they could pause or exit the simulation at any time without penalty. At the end of each session, a structured debriefing not only explained coping strategies but also reinforced participants’ ability to respond effectively to scams.

6 Results

Below, we present findings from quantitative results and qualitative analysis drawn from think-aloud protocols and semi-structured interviews. Quantitative measures showed significant improvements across all Protection Motivation Theory constructs, while qualitative analysis revealed how participants experienced and responded to personalized simulations. Our reporting proceeds in four parts: (1) participant characteristics, (2) learning experience with the system, (3) learning outcomes (shifts in deepfake knowledge, perceived threat, and coping efficacy) (4) areas for improvement identified through participant feedback.

6.1 Participants

Pre-survey scores indicated moderate digital literacy ($M = 3.76$, $SD = 0.79$, Median = 3.75, range = 2.75–5.00) and AI literacy ($M = 3.71$, $SD = 0.56$, Median = 3.78, range = 2.25–4.50), both measured on 5-point Likert scales. While all participants reported using smartphones in their daily lives, usage patterns varied from basic messaging to more advanced activities, including online banking and shopping. Despite moderate AI literacy scores, familiarity with AI as a general concept did not translate to understanding of deepfakes specifically. Only 5 of 21 participants reported having heard the term “deepfake” before the study, and most were learning about it for the first time in this study. Despite this limited prior exposure to deepfakes, all participants engaged actively with the study.

6.2 Learning Experience

Participants rated the overall learning experience positively across usability, satisfaction, and perceived usefulness. Deepfake realism received more varied ratings, while character immersion remained high (see Fig. 3, Table 2).

System usability was rated above average, with a mean score of 76 ($SD = 15.39$) on the System Usability Scale, where 68 is considered the benchmark for average usability [16], suggesting that the

Table 3: Descriptive statistics and Wilcoxon signed-rank test results for pre- and post-program Protection Motivation Theory constructs. All measures used 5-point Likert scales.

Construct		Pre-test		Post-test		α	Wilcoxon		
		<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>		<i>Z</i>	<i>p</i>	<i>r</i>
Knowledge		3.10	0.82	4.10	0.64	.89	3.148**	.002	.764
Perceived Threat	Perceived Vulnerability	2.95	0.88	3.86	0.76	.74	3.132**	.002	.760
	Perceived Severity	3.48	0.71	4.19	0.67	.62	2.588*	.010	.628
Coping Efficacy	Self-Efficacy	3.05	0.79	3.90	0.72	.83	2.926**	.003	.710
	Response Efficacy	3.24	0.74	4.14	0.60	.76	3.213**	.001	.779

* $p < .05$, ** $p < .01$. All effect sizes represent large effects ($r \geq .50$).

interface met usability expectations for our target group of older adults. Overall satisfaction with the learning experience was high (Median = 5.0, IQR = 0.70). Participants described the program as “well-structured and easy to follow” (P07) and noted that they could “complete it on [their] own without much difficulty” (P10).

Perceived usefulness was also highly valued (Median = 4.5, IQR = 1.0), with participants noting that the program felt directly relevant to situations they or their family members might encounter. One participant said, “I want to redo it at home—the scenarios looked too real” (P07). As such, several mentioned wanting to revisit the content or recommend it to relatives. This reflects participants’ perception that the platform provided actionable knowledge applicable to protecting themselves and others from deepfake scams.

Participants’ perceptions of how closely the deepfake simulations resembled them varied considerably (Median = 3.0, IQR = 1.0). Some found the scenarios highly believable, with one noting, “the scenarios looked too real” (P07). Others, however, identified imperfections that reduced the sense of self-resemblance. Visual artifacts undermined believability for some: “My face was swapped, but everyone else looked the same, so it was clearly fake” (P12). Voice quality also varied, with one participant noting, “The voice was about 70% similar” (P10), while others mentioned accent mismatches that made the simulations feel less like themselves. These mixed reactions indicate that while the platform achieved believability for some, technical fidelity in reproducing participants’ likenesses varied across individuals.

Despite varied perceptions of simulation realism, participants reported high engagement with the animated self-resembling guide (Median = 4.5, IQR = 1.0). The character’s resemblance to participants—featuring their own voice as narrator along with recognizable visual features such as hairstyle and glasses—helped maintain attention throughout the learning experience. Many participants emphasized that seeing and hearing themselves in the simulation heightened their engagement: “I focused more because it was my face and voice” (P13). This suggests that the familiar, personalized guide successfully served its intended role as a consistent presence that kept learners connected to the educational content.

6.3 Learning Outcomes

6.3.1 Knowledge of Deepfakes: Consolidating Fragmented Understanding. As shown in Fig. 4 (full statistics in Table 3), participants

reported significantly higher deepfake-related knowledge after completing the program ($Z = 3.148$, $p = .002$, $r = .764$). Interviews revealed how this improvement unfolded. For some, the session was their first encounter with the term *deepfake*, leading to reactions of surprise: “I didn’t know there were fake videos like that” (P20). For others, the program helped consolidate fragmented knowledge they had previously encountered through media or hearsay. These participants often described how the session clarified confusing terms or filled gaps in their understanding. As one explained, “I used to get all the terms mixed up, but now it’s clearer” (P10), while another reflected, “I definitely understand it better now than watching the [deepfake-related] videos” (P3).

Across both groups, participants highlighted the value of the scaffolded flow of the modules. The combination of short explanations, illustrative real-world examples, personalized simulations, and review quizzes made the unfamiliar technology easier to grasp and less overwhelming. As one noted, “The explanations were really easy to follow” (P2). These reflections suggest that the step-by-step progression might have supported participants in connecting new terminology to concrete scenarios, which in turn seemed to foster a clearer and more confident understanding.

6.3.2 Perceived Threat: Strengthening Awareness of Deepfake Risks. Survey results suggested increases in both dimensions of perceived threat after completing the program. Perceived vulnerability significantly increased ($Z = 3.132$, $p = .002$, $r = .760$), indicating that participants became more aware of their own potential exposure to scams. Perceived severity also significantly increased ($Z = 2.588$, $p = .010$, $r = .628$), which can be interpreted as a heightened sense of the broader societal consequences of deepfakes.

Interview responses reinforced these findings, particularly regarding personal vulnerability. Self-relevant simulations using participants’ own faces and voices were often described as shocking and realistic: “It was shocking to hear my voice saying someone had been in an accident” (P01); “I was so surprised... this could really happen to my son” (P06). These personalized scenarios made threats feel immediate and personally relevant. Some recalled acquaintances or relatives who had been victims of fraud, one participant shared, “I heard about someone who lost money to a voice phishing call. It’s happening all around us” (P15), reflecting a heightened sense of awareness.

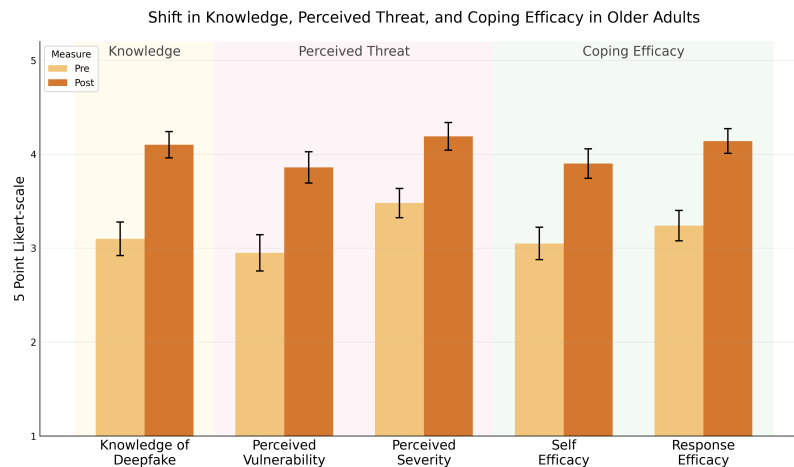


Figure 4: Pre- and post-survey comparisons across five constructs, grouped into three higher-order categories: *Knowledge of Deepfakes*, *Perceived Threat* (Vulnerability and Severity), and *Coping Efficacy* (Self- and Response-Efficacy). All measures showed improvement following the intervention.

Beyond personal vulnerability, participants also developed awareness of deepfakes' broader societal consequences. Several noted that the real-world examples presented in the learning modules, such as news articles about people falsely accused using deepfake images, helped them understand the wider implications of this technology. One participant noted that *"The news articles about people falsely accused using deepfake images really stayed with me"* (P01). This suggests that self-referential simulations heightened personal threat perception, where real-world examples supported a more comprehensive understanding of deepfake risks at both individual and societal levels.

Most participants reported increased threat awareness, while a small number expressed more cautious views. One participant suggested that highly realistic scenarios might overwhelm those with lower digital literacy: *"Some people might just stop using their phone out of fear—it could backfire"* (P11). Another participant with prior deepfake knowledge explained that the material did not feel particularly new and said *"I already knew about deepfakes, so it didn't feel that new"* (P12). These comments suggest that the impact of self-referential simulations may vary depending on participants' baseline familiarity with digital technologies.

6.3.3 Coping Efficacy: Building Confidence Through Scaffolded Practice. Analysis indicated significant increases in both self-reported self-efficacy and response efficacy. Self-efficacy increased significantly ($Z = 2.926$, $p = .003$, $r = .710$), and response efficacy also increased significantly ($Z = 3.213$, $p = .001$, $r = .779$).

Interview responses provided insight into these changes. Several participants described feeling more capable of identifying manipulation after the session. One noted, *"Now I'll be more skeptical when watching videos"* (P7), while another said, *"I think I can spot fakes by their gestures or speaking style"* (P20). Several participants also linked their growing confidence to the quiz component: *"At first, I felt lost, but after getting quiz answers right, I felt like I could actually do it"* (P5).

When asked how they would respond to suspicious content in the future, several participants described protective strategies they would employ. Some recalled protective strategies that had been introduced in the modules, such as verifying senders or calling back suspicious numbers, while others extended these lessons to articulate additional decision rules in their own words: *"If it's not from an official account, I won't trust it"* (P11); *"If something feels off, I'll ignore it"* (P14).

Taken together, these accounts suggest that participants were not only recalling strategies demonstrated in the program but also beginning to adapt or elaborate on them. This suggests that the program may have helped translate coping principles into practical decision-making rules, thereby reinforcing a sense of agency and practical readiness.

6.4 Areas for Improvement

While participants demonstrated significant gains in threat awareness and coping confidence, interviews revealed three areas where our design approach could be strengthened: bridging awareness to practiced readiness, adapting to diverse learning needs, and supporting peer-based learning.

6.4.1 Bridging Awareness to Practiced Readiness. Despite significant increases in self-efficacy and response efficacy, several participants noted a gap between conceptual understanding and practical readiness. While the program effectively heightened threat awareness, some felt they would benefit from practicing actual responses: *"I understood the explanation, but I'm not sure if I could actually follow those steps when I'm panicking during a real call"* (P14). Several participants explained specific practice needs. They mentioned wanting to practice verbal responses, such as what to say when receiving a suspicious call or how to challenge unexpected requests, rather than only learning strategies conceptually and practicing through quizzes. These reflections suggest that while the scaffolded learning steps effectively built awareness and introduced coping

strategies, incorporating interactive practice scenarios that allow learners to rehearse actual verbal responses in high-stakes scenarios could further strengthen the connection between knowledge and action.

6.4.2 Adapting Content to Varied Baseline Knowledge. Participants' responses revealed a fundamental tension in our one-size-fits-all approach. Participants' emotional responses to the scenarios varied depending on their baseline familiarity with digital threats. Some found the scenarios appropriately attention-grabbing. Others, however, expressed concern that the threat intensity might feel overwhelming. One participant described becoming overly skeptical: *"Now everything looks fake to me"* (P08). Conversely, those already familiar with deepfakes found the material insufficiently challenging.

This diversity in responses prompted participants to explicitly advocate for differentiated learning pathways. One emphasized the need for layered instruction: *"People unfamiliar with AI need step-by-step explanations. Mixing beginners and advanced learners in the same program doesn't work"* (P11). Meanwhile, more experienced participants expressed interest in advancing beyond foundational content: *"This was just the basics. I'd like to see more advanced content"* (P3). These responses suggest that a uniform approach risks overwhelming novices while under-stimulating experienced learners, pointing toward the need for adaptive pathways based on learners' prior knowledge.

6.4.3 Opportunities for Peer Learning and Social Sharing. Beyond individual learning, participants expressed a strong interest in sharing what they learned with family members. Notably, some recognized that personalized deepfakes could serve as compelling peer education materials: *"If I send my own deepfake video to a friend, they will definitely watch it. But if I just send a sample website, they probably won't look at it"* (P15). This insight suggests that self-referential artifacts may leverage existing trust relationships to overcome the low engagement typical of forwarded educational content.

Rather than viewing themselves solely as learners, several reframed their role as protectors of younger generations. Multiple participants expressed concern that children might be especially vulnerable to voice-based impersonation. As one explained, children lack the life experience needed to recognize manipulation: *"Young people can also fall for these scams. Especially if children hear their mother's voice telling them to follow someone, they would believe it. Children are more vulnerable. We have life experience to recognize suspicious situations, but children don't"* (P19). These reflections highlight opportunities for peer-based approaches that position older adults as active contributors to family cybersecurity, and suggest the value of supporting continuous learning through shareable resources or periodic updates about emerging threats.

7 Discussion

Our findings showed that *DeepAware* helped older adults not only improve their understanding of deepfakes but also report heightened perceptions of risk and greater confidence in their ability to respond. Participants emphasized that the system's personalized simulation and scaffolded learning experience made threats

feel personally relevant and tangible while also providing concrete strategies for action.

DeepAware was designed to address two recurring challenges identified in both in prior studies and in our formative study with digital educators. First, older adults often struggled to perceive these scams as personally relevant; abstract explanations or news stories rarely conveyed why they mattered for everyday safety [18, 23, 55]. Second, even when learners were attentive, educators noted that it was difficult to provide concrete strategies for how to respond, leading safety instruction to rely mainly on simple avoidance rules. These insights echo prior research in cybersecurity education, which shows that knowledge-based instruction can raise awareness but does not consistently translate into protective action [18, 23, 55].

To address these gaps, we examined our design approach on how personalized experiential learning could support older adults in recognizing and responding to deepfake scams. Guided by experiential learning principles [45, 56, 72, 107], we designed self-relevant simulations by embedding users' faces and voices to heighten perceived threat and coping efficacy.

Findings from our evaluation suggest that personalized experiential simulation can be a promising approach for raising both threat perception and coping efficacy. Prior work has proposed experiential approaches to digital security education, such as experiential dark-pattern training [108] and card game-based activities for practicing security decisions [17]. Researchers have also investigated deepfake-scam awareness through a participatory workshop with older adults [109]. In parallel, personalization studies have examined how to adapt security and privacy guidance to individuals' motivations [34], personas [53], and the diverse needs of older adults [74]. These approaches train learners to recognize and respond to threats, but the threats themselves remain generic, where learners practice detecting scams that could target anyone. Our approach shifts this relationship by casting learners themselves as the target, showing participants their own face manipulated in fake news and letting them hear their own voice used to deceive family members. This direct embodiment of victimhood, as opposed to merely observing others' experiences, may explain why participants reported that threats felt "real" and "immediate" in ways that prior warnings had not.

This finding has implications beyond deepfake education. Simulations that reveal how an individual's identity could be exploited, instead of relying on generic threat cues, may improve risk communication in areas where risks feel abstract, such as phishing, health misinformation, and financial fraud. By shifting learners from observers to targets, experiential systems can transform "this could happen to someone" into "this is happening to me," potentially strengthening both threat appraisal and protective motivation.

However, self-referential personalization did not uniformly empower all participants. While many found seeing themselves in scam scenarios increased clarity and engagement, a minority reported feeling overwhelmed by the realism. Some questioned whether they could trust any digital communication afterward, while some with prior familiarity with deepfakes found the content less engaging. These divergent reactions highlight that self-relevance can strengthen protective reasoning for some learners yet trigger cognitive overload or disengagement for others. When emotional arousal

exceeds available cognitive resources, learners may struggle to engage in reflective evaluation and instead rely on fast, defensive reactions [12]. Fear-appeal research similarly shows that high-threat messages can backfire when threat appraisal is not matched with adequate coping efficacy, resulting in avoidance rather than protective action [88, 98]. These findings suggest that effective personalization depends on aligning simulation intensity with learners' baseline knowledge, emotional readiness, and digital familiarity.

7.1 Design Implications for Simulation-Based Cybersecurity Education with Older Adults

Based on our formative findings and system evaluation, we outline design implications for incorporating self-relevant simulation into cybersecurity education for older adults.

7.1.1 Design for Learner Diversity. Our findings revealed substantial variation in older adults' prior exposure to deepfakes. Some participants encountered the concept for the first time, whereas others were already familiar with deepfakes and found deepfake scenarios less engaging. DeepAware's uniform approach could not accommodate this diversity, as the same simulation that heightened awareness for novice learners felt redundant to experienced learners, while scenarios that engaged experienced learners risked overwhelming those new to the concept.

To address this variability, future simulation-based education should calibrate scenario intensity to different learner profiles. For learners with limited prior exposure, early scenarios may need to begin with lower-pressure scenarios that introduce threat concepts gradually. Starting with less emotionally charged scenarios reduces cognitive load and allows learners to focus on recognizing risk cues without being overwhelmed [58, 96]. Progressively increasing intensity helps novices build confidence before encountering high-pressure scenarios.

Learners with more experience, by contrast, may benefit from scenarios that introduce greater complexity, such as romance fraud, investment fraud, or government impersonation [30, 77]. While *DeepAware* focused on identity theft and fake news as entry points, expanding the scenario repertoire would help maintain engagement across different levels of prior knowledge while ensuring that simulations remain appropriately challenging [92, 97].

7.1.2 Scaffold Coping Efficacy through Simulation-Based Rehearsal. Our findings indicated that older adults responded most positively when coping strategies were simple and actionable, such as hanging up and calling back or using a code word with family. However, our current design stopped short of enabling learners to practice these strategies in realistic contexts. While the scaffolded learning stages introduced coping strategies conceptually and reinforced them through quizzes, participants noted that they lacked opportunities to rehearse actual verbal responses, such as what to say when receiving a suspicious call or how to challenge unexpected requests.

To bridge this gap between knowledge and action, future systems might extend this approach by integrating more immersive rehearsals, such as simulated deepfake scam calls. In these scenarios, learners could receive an audio message in a familiar voice—mimicking a relative, for example—and be prompted to decide whether to trust

it, call back, or verify with a shared family passcode. Immediate feedback in these cases can reinforce correct responses while reframing errors as learning opportunities. Prior work on phishing education shows that such scenario-based inoculation fosters durable “pause–reflect–act” habits in high-pressure situations [59], and that interactive simulations, including vishing-specific trainings, help users practice protective responses [20, 42, 102].

In this way, simulation not only strengthens coping efficacy—but also builds self-efficacy through repeated, low-stakes success experiences. For participants less familiar with scams, this rehearsal can reduce anxiety by showing that effective responses are within reach; for those with prior knowledge, it offers opportunities to refine strategies and apply them across varied contexts. Together, these features highlight how simulation can scaffold coping strategies in ways that are emotionally resonant, practical, and sustainable.

7.1.3 Support Self-Directed and Peer-Driven Learning. Participants frequently expressed a desire to revisit the materials after the session, repeat modules, or share what they had learned with family members. These reflections suggest that older adults value opportunities to review, rehearse, and reinforce content at their own pace rather than relying on one-time instruction. Simulation-based education can support this need by offering short, modular scenarios that are easily revisited and by incorporating features such as audio narration and visual cues, which participants identified as making the content more approachable.

Our findings also revealed that learning often extended into social contexts. Many noted that personalized deepfake artifacts could effectively engage friends and family, as people are more likely to pay attention to content featuring someone they personally know. These interactions illustrate how simulation can spark conversation and encourage collaborative verification practices, reinforcing protective behaviors beyond the individual learner. Designing family-oriented scenarios or shareable modules could build on this tendency, enabling learners to share selected scenarios with family or friends [60] and potentially overcoming the low engagement typically associated with forwarded cybersecurity content [83].

7.2 Limitations and Future Work

While our study provides preliminary evidence that personalized, simulation-based education can enhance older adults' awareness of deepfakes and their confidence in coping with them, the findings should be interpreted in light of several limitations that also point to directions for future work.

First, our sample reflects a specific population and context. All participants were older adults residing in urban areas of South Korea who were already active smartphone users. Recruitment through digital educators may have selected for participants with higher baseline digital literacy, contributing to ceiling effects in some measures. It remains unclear whether older adults with limited digital exposure would benefit equally or require additional scaffolding. Future work should recruit more diverse samples across different levels of digital literacy, geographic contexts, and cultural backgrounds to examine the generalizability of personalized simulation approaches.

Second, we adopted a single-group pre–post design without a control condition. While this design was a feasible first step in exploring the preliminary outcomes of a new design approach, it limits causal inference. The observed improvements in knowledge, threat perception, and coping efficacy should be interpreted as preliminary evidence rather than definitive proof of effectiveness. These gains may partially reflect general learning effects from repeated exposure to deepfake concepts or survey items, independent of the system’s specific design features. Moreover, because our intervention foregrounds participants’ own faces and voices, it is possible that some responses reflect privacy-priming effects [4, 93]. Future work should employ randomized controlled trials to rigorously compare simulation-based interventions with alternative formats such as lecture-based workshops or public service campaigns, providing stronger evidence of relative effectiveness.

Third, demand characteristics and social desirability may have influenced participants’ responses [32, 81]. Participants who invested time, shared personal photos, and engaged with researchers may have felt inclined to report positive outcomes [25]. We attempted to mitigate these concerns by explicitly probing for difficulties, areas for improvement, and aspects that did not work well during the interviews. As a result, participants could freely identify limitations and suggested improvements, as reported in Section 6.4. Nonetheless, the findings may not fully capture the range of participant experiences.

Fourth, the study focused on short-term outcomes following a single 90-minute session. This approach captured immediate changes in knowledge, perceived threat, and coping efficacy with high measurement consistency. Nevertheless, it remains unclear whether these gains persist over time or translate into real-world behavioral change. Future longitudinal research should investigate the retention of threat appraisal, sustained engagement, and actual scam-resisting behaviors over extended periods.

Finally, the evaluation focused on two modules—identity theft and fake news, representing common but not exhaustive deepfake scam tactics. The video generation tools produced convincing facial animations but offered limited body movement and background realism, constraining scenario immersion. Future work should explore improved generation techniques and expand scenarios to include romance fraud, investment schemes, and government impersonation, providing more comprehensive coverage of the evolving deepfake threat landscape.

8 Conclusion

This paper introduced *DeepAware*, a simulation-based learning platform designed to help older adults recognize and respond to deepfake scams. Grounded in Protection Motivation Theory and experiential learning, the system embeds learners’ own photos and voices into self-relevant identity theft and fake news scenarios and guides them through scaffolded stages, from concept introduction to coping strategies. Findings from a mixed-method evaluation with 21 older adults revealed improvements in knowledge, perceived threat, and coping efficacy. Participants described the personalized simulations as engaging and realistic, often noting that the simulation provided a safe space to understand threats and practice actionable responses.

Building on the findings of our formative study and user evaluation of *DeepAware*, we identify three design implications for future systems. First, simulation-based interventions should calibrate scenario difficulty and diversify content to match learners’ digital skills and prior experience, keeping scenarios appropriately challenging yet accessible. Second, systems should embed safe opportunities to rehearse coping strategies, enabling learners to build confidence through practice. Third, platforms should support sustained, self-directed engagement by allowing learners to revisit modules at their own pace and share them with peers or family members. Together, these implications highlight the promise of experiential, self-relevant learning to help older adults move from recognizing threats to taking protective action, fostering confidence and resilience in digital security. We hope this work contributes to the broader effort to design inclusive and impactful educational interventions that empower older adults to recognize and respond to emerging AI-driven threats.

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A Formative Study Interview Protocol

We organized our semi-structured interview protocol into three areas. Interviewers selected follow-up questions based on participants' responses.

1. *Current Practices and Challenges in Teaching Older Adults.*

- What kinds of digital skills or programs do you currently teach to older adults?
- What aspects of smartphone or internet use do older adults find most difficult?
- What are some common misconceptions or breakdowns you observe across learners?
- How do you typically structure lessons to support confidence and reduce anxiety?
- Are there topics that you want to teach but find difficult to cover within your current curriculum?

2. *Explaining Digital Safety and Emerging Risks (e.g., Deepfakes).*

- How do you currently explain online risks (e.g., phishing, scams, impersonation)?
- Have you discussed AI-generated media or deepfakes in your classes? If so, how?
- What examples or materials do learners respond to most strongly when discussing digital safety?
- What challenges do you face when introducing new or unfamiliar threats like deepfakes?
- In your experience, what makes safety instruction more understandable for older adults?

3. *Learners' Reactions to Risk Scenarios and Coping Strategies.*

- Can you recall moments when older adults became noticeably more cautious or alert to scams?
- What kinds of scenarios or stories capture their attention most effectively?
- How do older adults typically react to warnings or risk demonstrations?
- What coping strategies (e.g., verification steps, asking questions) seem easiest for them to adopt?
- What forms of practice, feedback, or guidance help reduce fear while encouraging safe behavior?

B User Study Participants Information

Table 4 presents detailed demographic information for all 21 participants.

Table 4: Participants Demographics

ID	Age	Gender	Occupation	Digital devices in use
P1	62	Male	Employed	Smartphone, Computer, Smart Watch, Smart TV
P2	72	Male	Employed	Smartphone
P3	68	Female	Retired	Smartphone
P4	69	Female	Retired	Smartphone
P5	67	Male	Employed	Smartphone, Smart Watch
P6	68	Female	Retired	Smartphone, Smart TV
P7	80	Female	Retired	Smartphone
P8	63	Female	Employed	Smartphone
P9	63	Female	Employed	Smartphone
P10	63	Female	Employed	Smartphone
P11	63	Male	Employed	Smartphone, Computer
P12	68	Male	Employed	Smartphone, Tablet, Computer, Smart Watch, Smart TV
P13	63	Female	Retired	Smartphone, Tablet, Computer, Smart Watch, Smart TV
P14	62	Female	Employed	Smartphone, Computer, Smart Watch, Smart TV
P15	63	Female	Retired	Smartphone, Computer
P16	70	Female	Retired	Smartphone, Computer
P17	61	Female	Retired	Smartphone, Computer
P18	66	Female	Retired	Smartphone, Computer, Smart Watch, Smart TV
P19	65	Female	Retired	Smartphone
P20	62	Female	Retired	Smartphone
P21	63	Male	Retired	Smartphone

C Learning Experience Measurement Items

Table 5 presents the survey items used to assess participants’ learning experiences, including satisfaction, perceived usefulness, deepfake realism, and character immersion.

Table 5: Survey Items for Learning Experience Constructs

Construct	Survey Items	Cronbach’s α
Satisfaction	The overall educational structure (explanation → examples → practice → alternatives → wrap-up) effectively supported my learning.	.77
	Organizing the content into two separate practice topics (fake news and identity theft) made the learning manageable.	
	The two practice sessions (fake news, identity theft) maintained my engagement throughout the program.	
Perceived Usefulness	The strategies I learned from both practices are applicable to situations I may encounter in real life.	.86
	The fake news practice equipped me with practical skills to distinguish false information.	
Deepfake Realism	The AI-generated voice and character accurately reproduced my actual voice and appearance.	–
Character Immersion	The AI character created a sense of personalized, direct communication with me.	.91
	My AI character maintained my attention and involvement throughout the educational content.	

– indicates single-item measure.

D PMT Construct Measurement Items

Table 6 presents the survey items used to measure Protection Motivation Theory constructs and deepfake knowledge.

Table 6: Survey Items for PMT Constructs and Deepfake Knowledge

Construct	Survey Items	Cronbach's α
Knowledge of Deepfake	I know the concept and definition of deepfakes. I recognize that deepfake technology can manipulate faces and voices. I understand that deepfake technology is becoming increasingly sophisticated.	.89
Perceived Vulnerability	I am aware that someone could send a deepfake video to deceive me or my family. I believe I could be a target of deepfake scams. I think deepfakes could be used to impersonate me or people I know.	.74
Perceived Severity	I believe deepfakes can contribute to the spread of false information in society. I think deepfake scams can cause serious harm to individuals and families.	.62
Self-Efficacy	I can tell if the news or videos I see online have been manipulated. I can tell whether the person I interact with online is human or AI.	.83
Response Efficacy	When I see shocking or surprising videos, I will stop and think before believing them. I believe verification strategies (such as calling back or using a family code word) can effectively prevent deepfake scams. I think the coping strategies I learned are practical and useful for protecting myself.	.76

E User Evaluation Interview Protocol

We organized our semi-structured interview protocol into four areas. Interviewers selected follow-up questions based on participants' responses.

1. *Emotional Engagement and Self-Relevance.*

- How did you feel when you saw your own face and voice in the simulation scenarios?
- Was it surprising, scary, or just interesting? Please describe your feelings.
- What was your experience with the animated self-resembling character that guided you through the program?
- How did the character affect your engagement with the program, if at all?
- How would you compare using your own character and voice versus a generic announcer or teacher character?

2. *Changes in Perceived Threat and Understanding.*

- How clear were the explanations of deepfake and deepvoice concepts?
- Were there any parts that were difficult to understand or that you wish had been explained differently?
- After completing the program, has your view of deepfake scams changed? If so, how?

- Which moments stood out to you, and why?
- How did you feel about the level of detail and intensity in the scenarios?

3. *Applicability of Coping Strategies and Learning Transfer.*

- What did you think of the quizzes presented during the program?
- How do you think the quizzes affected your understanding of how to respond in real situations?
- After this program, how do you think you would respond if you encountered a suspicious video?
- Can you describe any specific actions you might take?
- How realistic or practical do you think these coping strategies are for your daily life?

4. *Overall Experience and Suggestions for Improvement.*

- What aspects of the program worked well for you, and what did not?
- Would you want to revisit this content or share it with others? Why or why not?
- What would you change about the program if you could?
- Were there any aspects that were confusing, frustrating, or difficult to navigate?